Shazil Shiraj Katchhi

M.Sc. AI Student | Machine Learning, Deep Learning & Embedded AI | Python & IoT Systems

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PROFESSIONAL SUMMARY

M.Sc. Artificial Intelligence student at BTU Cottbus-Senftenberg with practical experience in machine learning, deep learning, IoT systems and embedded hardware development. IEEE-published researcher who has built and deployed real AI systems end-to-end - from hardware integration and dataset creation through model training, evaluation and production deployment. Smart India Hackathon national finalist (Top 5). Motivated by research, innovation and applied AI that creates measurable real-world impact. Seeking internship, working student or research assistant roles in AI, ML, Data Science or Software Development across Germany.

EDUCATION

M.Sc. Artificial Intelligence Mar 2026 - Present

BTU Cottbus-Senftenberg, Germany

Coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Data Analytics, Software Engineering, AI Systems

B.E. Computer Science & Engineering (AI & ML) 2021 - 2025

University of Mumbai, India

Coursework: ML, Deep Learning, Algorithms, Embedded Systems, Data Structures, Databases, Signal Processing

PROFESSIONAL EXPERIENCE

IoT & Database Intern Feb 2024 - May 2024

Skmotiv, Mumbai, India

- Designed and prototyped a Raspberry Pi-based IoT module for fuel tanker fleet management, capturing real-time data: GPS location, fuel capacity, fuel type and truck weight before and after fuelling.
- Implemented a structured database schema (SQLite) to store, query and report fleet telemetry data; wrote Python scripts running on embedded Linux for data acquisition and logging.
- Integrated hardware sensors with the Raspberry Pi GPIO interface; tested the system under real operating conditions and documented the full pipeline for handover.

RESEARCH PUBLICATION

A Deep Learning Framework for Sign-to-Speech Translation Mar 2025

IEEE Xplore - ESCI 2025, Co-Author

- Designed a multimodal deep learning pipeline combining CNN (spatial feature extraction), Vision Transformer - ViT (global attention) and LSTM (temporal modelling) with MediaPipe landmark detection.
- Created and annotated the evaluation dataset; implemented systematic benchmarking with per-class precision, recall and F1-score across all sign classes.
- Deployed real-time inference on Raspberry Pi 4 embedded hardware running Linux; resolved latency bottlenecks by parallelising audio and video processing threads. Achieved 91% accuracy.

PROJECTS

Sign Language to Speech - Smart India Hackathon Top 5 National Dec 2024 - Mar 2025

GitHub

Tech: Python, PyTorch, MediaPipe, CNN, ViT, LSTM, Raspberry Pi 4, Linux, OpenCV

- Built a wearable AI system where a Raspberry Pi embedded in smart glasses captures real-world visual input, runs a multimodal deep learning pipeline and produces speech output in real time.
- Optimised for embedded hardware constraints; resolved latency via parallel threading. Presented and validated against thousands of competing teams - Top 5 nationally.

Real-Time Object Detection & Benchmarking GitHub 2024

Tech: Python, TensorFlow, OpenCV, NumPy

- Built a real-time object detection system; benchmarked performance across model configurations with systematic evaluation of accuracy and inference speed.

Credit Card Fraud Detection - ML Classification Pipeline GitHub 2024

Tech: Python, Scikit-learn, Pandas, NumPy, Matplotlib, SQL

• End-to-end ML pipeline: data ingestion, cleaning, feature engineering, model training (classification) and evaluation with precision, recall and F1-score reporting.

Generative AI - Fashion GAN GitHub 2024

Tech: Python, TensorFlow, Keras, NumPy, Matplotlib

• Trained a Generative Adversarial Network (GAN) on the Fashion MNIST dataset; implemented custom Generator and Discriminator architectures and tracked training loss convergence.

TECHNICAL SKILLS

ML & Deep Learning	Python, PyTorch, TensorFlow, Keras, Scikit-learn, CNN, ViT, LSTM, GAN, NLP, Computer Vision, MediaPipe, model evaluation
IoT & Embedded	Raspberry Pi 4 (Linux), GPIO integration, sensor data acquisition, real-time pipelines, SQLite, embedded Python
Data & Analytics	Pandas, NumPy, Matplotlib, OpenCV, SQL, data cleaning, feature engineering, statistical analysis
Dev & Tools	Git, GitHub, Jupyter Notebook, Docker (basic), REST APIs (basic), Linux, Bash, VS Code
Languages	English C2 (IELTS 8.5), German A2 (progressing), Hindi Native, Urdu Native

LEADERSHIP & ACTIVITIES

• **Chairperson** - IEEE MHSSCE Student Chapter 2024 - 2025

• **General Secretary** - SACE Committee, MHSSCE Jan - Jun 2025

• **Smart India Hackathon Finalist** - Top 5 National - Government of India Dec 2024 - Mar 2025

• **IEEE Co-Author** - ESCI 2025 - Sign-to-Speech Deep Learning Framework Mar 2025

ADDITIONAL INFORMATION

Availability	Werkstudent (up to 20 hrs/week), Pflichtpraktikum or research assistant. Germany-wide. Immediately available.
Work Auth	Enrolled M.Sc. student in Germany. Eligible to work as Werkstudent or intern.